

Homework 12 Solutions

Donald Timpson

September 16, 2022

Problem 26.1

- a) 1) $g \circ f, f \circ g$
2) $g \circ f = \{(1, 1), (2, 1), (3, 1)\}, f \circ g = \{(2, 2), (3, 2), (4, 2)\}$
3) $g \circ f \neq f \circ g$
- b) 1) $g \circ f, f \circ g$
2) $g \circ f = \{(1, 1), (2, 2), (3, 3)\}, f \circ g = \{(2, 2), (3, 3), (4, 4)\}$
3) $g \circ f \neq f \circ g$
- c) 1) $g \circ f$, Undefined.
2) $g \circ f = \{(1, 1), (2, 5), (3, 3)\}$
- d) 1) $g \circ f, f \circ g$
2) $g \circ f = \{(1, 1), (2, 1), (3, 4), (4, 1)\}, f \circ g = \{(1, 4), (2, 4), (3, 1), (4, 1)\}$
3) $g \circ f \neq f \circ g$
- e) 1) $g \circ f, f \circ g$
2) $g \circ f = \{(1, 4), (2, 5), (3, 1), (4, 2), (5, 3)\}, f \circ g = \{(1, 4), (2, 5), (3, 1), (4, 2), (5, 3)\}$
3) $g \circ f = f \circ g$
- f) 1) $g \circ f, f \circ g$
2) $g \circ f = (x^2 - 1)^2 + 1, f \circ g = (x^2 + 1)^2 - 1$
3) $g \circ f \neq f \circ g$
- g) 1) $g \circ f, f \circ g$
2) $g \circ f = x - 4, f \circ g = x - 4$
3) $g \circ f = f \circ g$
- h) 1) $g \circ f, f \circ g$
2) $g \circ f = x + 2, f \circ g = x - 1$
3) $g \circ f \neq f \circ g$
- i) 1) $g \circ f$, Undefined.
2) $g \circ f = \frac{1}{x} + 1$ for all $x \in \mathbb{Q}$.
- j) 1) $g \circ f$, Undefined.
2) $g \circ f = \{(a, b) : (a, c) \in id_A, (c, b) \in id_B\}$

Problem 26.5

Proof. If A and B are sets and $f : A \rightarrow B$, then $f \circ id_A = id_B \circ f = f$.

1. Let A and B be sets and let $f : A \rightarrow B$.
2. Then $\text{dom}(f \circ id_A) = \text{dom}(id_A) = A$, $\text{dom}(id_B \circ f) = \text{dom}(f) = A$, and $\text{dom}(f) = A$.
3. Therefore $f \circ id_A$, $id_B \circ f$, and f all have the same domain, namely A .
4. Let $x \in A$.
5. Then $(f \circ id_A)(x) = f(id_A(x)) = f(x)$, $(id_B \circ f)(x) = id_B(f(x)) = f(x)$, and $f(x) = f(x)$.
6. Therefore $f \circ id_A$, $id_B \circ f$, and f all map x to the same value, namely $f(x)$.
7. Therefore $f \circ id_A = id_B \circ f = f$. □

Problem 26.6

Proof. If A and B are sets and $f : A \rightarrow B$ is one-to-one and onto, then $f \circ f^{-1} = id_B$ and $f^{-1} \circ f = id_A$.

1. Let A and B be sets and let $f : A \rightarrow B$ be one-to-one and onto.
2. Then $\text{dom}(f \circ f^{-1}) = \text{dom}(f^{-1}) = B = \text{dom}(id_B)$, and $\text{dom}(f^{-1} \circ f) = \text{dom}(f) = A = \text{dom}(id_A)$.
3. Therefore $f \circ f^{-1}$ and id_B have the same domain B , and $f^{-1} \circ f$ and id_A have the same domain A .
4. Let $y \in B$ and observe that since f is onto then $\exists x \in A$ such that $f(x) = y$.
5. Also, since f is one-to-one we know that $f^{-1}(y) = x$.
6. Then $(f \circ f^{-1})(y) = f(f^{-1}(y)) = f(x) = y$ and $id_B(y) = y$, showing that $f \circ f^{-1}$ and id_B map y to the same value.
7. Therefore $f \circ f^{-1} = id_B$.
8. Now, let $x \in A$ and let $y = f(x)$.
9. Observe that since f is one-to-one then x is the only value that maps to y , so $f^{-1}(y) = x$.
10. Then $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(y) = x$ and $id_A(x) = x$, showing that $f^{-1} \circ f$ and id_A both map x to x .
11. Therefore $f^{-1} \circ f = id_A$.
12. Therefore $f \circ f^{-1} = id_B$ and $f^{-1} \circ f = id_A$. □

Problem 26.9

Let $f : A \rightarrow B$, and $g : B \rightarrow C$

Proof. If f and g are one-to-one, so is $g \circ f$.

1. Let $g(f(x)) = g(f(y))$.
2. Since g is one-to-one, then $f(x) = f(y)$.
3. Since f is one-to-one, then $x = y$.
4. Therefore $g \circ f$ is one-to-one. □

Proof. If f and g are onto, so is $g \circ f$.

1. Since f is onto, $\text{im } f = B$.
2. Since $\text{dom } g(f(x)) = \text{im } f$, then $\text{dom } g(f(x)) = B$.
3. Since g is onto, $\text{im } g(f(x)) = C$.
4. Therefore $g \circ f$ is onto. □

Proof. If f and g are bijections, so is $g \circ f$.

1. Since f and g are one-to-one, so is $g \circ f$ (proven above).
2. Since f and g are onto, so is $g \circ f$ (proven above).
3. Therefore $g \circ f$ is a bijection. □

Problem 27.2

- a) $\sigma = (1, 2, 4)(3, 6, 5)$
- b) $\pi = (1, 2, 3, 4, 5, 6)$
- c) $\pi \circ \pi = (1, 3, 5)(2, 4, 6)$
- d) $\pi^{-1} = (1, 6, 5, 4, 3, 2)$

e) $\iota = (1)(2)(3)(4)(5)$

f) $(1, 5, 4, 3, 2)$

g) $(1, 2, 6)(3, 5)(4)$

Problem 27.5

Let $\pi = (1)(2, 3, 4, 5)(6, 7, 8, 9)$, $\sigma = (1, 3, 5, 7, 9, 2, 4, 6, 8)$, and $\tau = (1, 9)(2, 8)(3, 5)(4, 6)(7)$.

a) $\pi \circ \sigma = (1, 4, 7, 6, 9, 3, 2, 5, 8)$

b) $\sigma \circ \pi = (1, 3, 6, 9, 8, 2, 5, 4, 7)$

c) $\pi \circ \pi = (1)(2, 4)(3, 5)(6, 8)(7, 9)$

d) $\pi^{-1} = (1)(2, 5, 4, 3)(6, 9, 8, 7)$

e) $\sigma^{-1} = (1, 8, 6, 5, 2, 9, 7, 5, 3)$

f) $\tau \circ \tau = (1)(2)(3)(4)(5)(6)(7)(8)(9)$

g) $\tau^{-1} = \tau$

Problem 27.6

Proof. It is not the case that $\forall \pi, \sigma \in S_n, \pi \circ \sigma = \sigma \circ \pi$.

1. Let $\pi = (1, 2, 3)$ and $\sigma = (1, 2)(3)$.

2. Then $\pi \circ \sigma = (1, 3)(2) \neq (1)(2, 3) = \sigma \circ \pi$. □

Problem 27.7

Proof. It is not the case that if π and σ are transpositions, then $\pi \circ \sigma = \sigma \circ \pi$.

1. Let $\pi = (1)(2, 3)$ and $\sigma = (1, 2)(3)$.

2. Then $\pi \circ \sigma = (1, 3, 2) \neq (1, 2, 3) = \sigma \circ \pi$. □